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Experiment Title: Design Star and snowflake schema

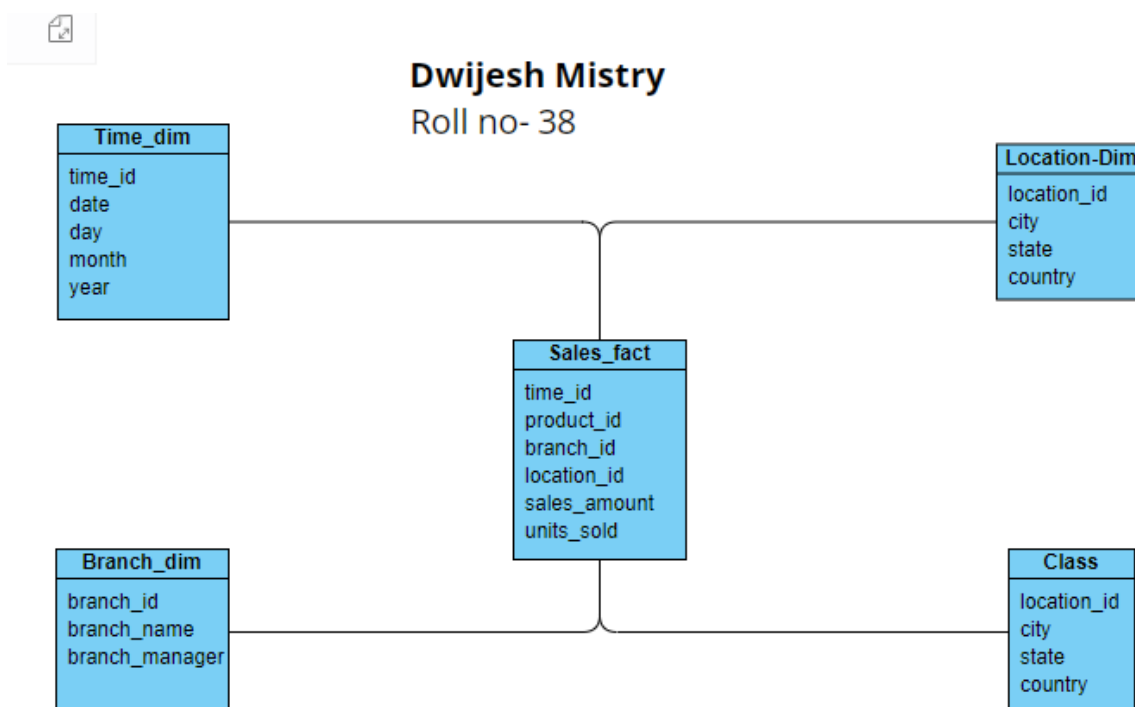
EXPERIMENT – 1

Problem statement:

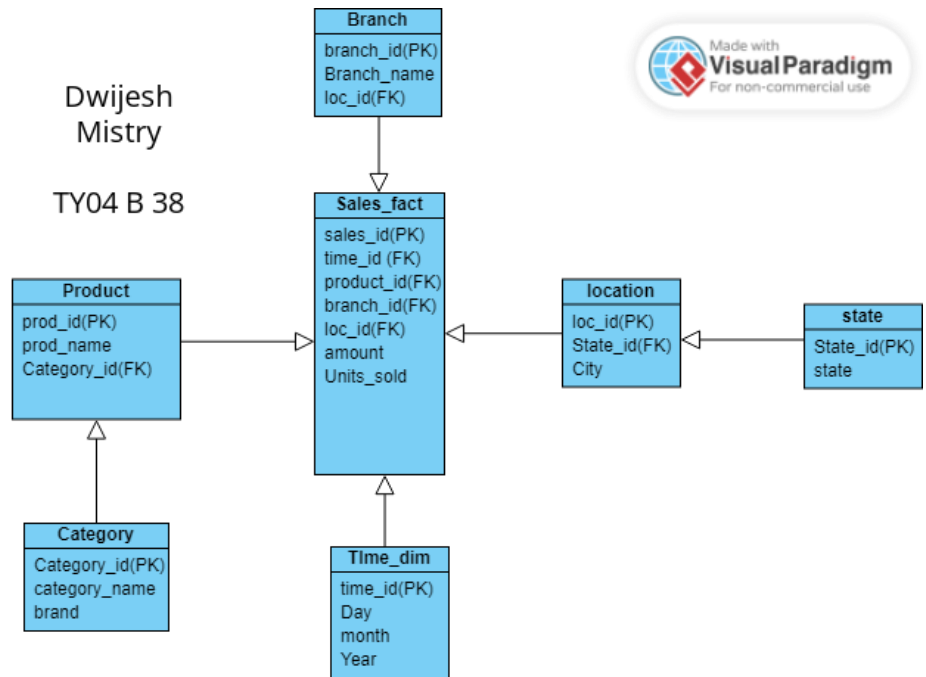
Electronics company has sales department. Consider 4 dimensions - Time, Product, Branch and location. The schema contains a central fact table sales with two measure sold and units sold

Output Diagram:

1. Star Schema



2. Snowflake Schema:



Review Questions:

1. In a star schema, how is the fact table typically related to the dimension tables?

Ans:

- In a star schema, the fact table sits at the center of the database design and stores quantitative, measurable data such as sales amount, quantity sold, or revenue.
- Each record in the fact table represents a business event (for example, a sale or transaction).
- The fact table contains foreign keys that reference the primary keys of the dimension tables.
- Each dimension table describes the context of the facts, such as time, product, customer, or location.
- These relationships are typically **one-to-many**, meaning one dimension record (e.g., a specific product) can be associated with many fact records (e.g., many sales of that product).
- Dimension tables are not related to each other; they connect only through the fact table, forming a star-like structure.

2. What is the main difference between a star schema and a snowflake schema?

Ans:

- The key difference lies in how dimension data is structured and normalized.
- In a star schema, dimension tables are denormalized, meaning all related attributes are stored in a single table. This makes the design simple and easy to understand, and queries run faster because fewer joins are required. However, it may lead to some data redundancy.
- In contrast, a snowflake schema normalizes dimension tables into multiple related tables. For example, a product dimension might be split into separate tables for product, brand, and category. This reduces data redundancy and saves storage space but results in a more complex schema and slower query performance due to additional joins.
- Overall, star schemas prioritize simplicity and query performance, while snowflake schemas prioritize normalization and storage efficiency.